Ananmay Som Singh

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EDUCATION

University of Maryland, College Park | College Park, MD **Expected May 2028**

Bachelor of Science in Computer Engineering

Relevant Coursework: Algorithms, Data Structures & OOP (Java), Computer Systems (C), Computer Organization, Compilers, Probability & Statistics

EXPERIENCE

GSAlpha Labs | Real Estate Transaction Automation **May 2026 – Present**

Software Engineering Intern · Part-time during academic year *San Francisco, CA*

- Built HomeFlow AI, a production **Next.js/TypeScript** platform automating the paperwork transaction coordinators charge **$350–600 per deal** for.
- Cut purchase-agreement data entry from **35 min to under 30 s**, at 96% field accuracy on 15 real deals, by building an **LLM** extraction pipeline.
- Let agents draft a purchase agreement from a 175-page disclosure packet in **under 5 minutes** by auto-filling fields and flagging only conflicts.
- Built source-grounded validation that checks each **LLM**-extracted fact against the PDF page it cites and flags mismatches for agent review.

theconviction.ai | Research Platform for Retail Traders **May 2025 – Aug 2025**

Software Engineering Intern *Remote*

- Built a **Python** ingestion pipeline aggregating SEC filings, earnings transcripts, and news on any stock for a retail-trader research platform.
- Loaded **2,491** filings, transcripts, and articles in one **8-minute** refresh by building a **FastAPI/PostgreSQL** pipeline with retries and dedup.
- Let traders verify every AI-extracted event against its source filing by building a **React** timeline that blocks uncited material events.

SEDS at UMD | SatFab CubeSat GPS Team **Sep 2024 – Feb 2026**

Software Engineer *College Park, MD*

- Built the **Python** test infrastructure for a CubeSat GPS receiver's 26 hardware modules; became the team standard after I trained 6 engineers.
- Cut the regression suite from **4 hours to 95 minutes** by running tests in parallel, turning an overnight job into a check before every merge.
- Replaced manual waveform review with automated pass/fail checks and reports, catching integration bugs in simulation before they reached hardware.

PROJECTS

Tape: Market Data Capture & Replay | *Go, AWS S3, ECS, CloudWatch, Terraform, Docker* AnanmayS/tape

- Built a concurrent **Go** market-data capture and replay service, defining **AWS** ECS infrastructure in **Terraform** with **GitHub Actions** CI.
- Handled **49.8k messages/sec** in load tests and cut storage **5.2x** by writing feeds as delta-encoded columnar files, replayable byte-for-byte.
- Caught **all 3 feed gaps** (4,415 missing messages) in fault tests by tracking sequence numbers, flagging bad data before any backtest used it.

Wildebeest: Distributed Wildlife Photo Pipeline | *TypeScript, Node.js, Python, PostgreSQL, Redis, Docker* AnanmayS/wildebeest

- Built a fault-tolerant distributed pipeline running **PyTorch** models (MegaDetector, SpeciesNet) on wildlife photos across CPU and GPU workers.
- Achieved **0 lost or duplicate results** across 30 forced worker crashes by building crash-safe task processing on **PostgreSQL** leases and fencing tokens.
- Reduced worker crash recovery time by **97%** (5.6 s to 0.16 s) using **Docker** events instead of heartbeat timeouts to detect dead workers.

Trading Arena: Futures Exchange Game | *C++, WebAssembly, TypeScript, React, CMake, GitHub Actions* AnanmayS/trading_arena

- Built a browser futures-trading game in **C++** compiled to **WebAssembly**, where players trade against 32 bots on a live order book.
- Matched **10.4M mixed ops/sec** (52 ns median) with a price-time order book that never allocates on the hot path, using pooled orders and bitsets.
- Kept native and **WASM** builds **byte-identical** with an integer-only engine checked in CI, and held P&L zero-sum across **250k** random operations.

TECHNICAL SKILLS

Languages: Python, Go, Java, C++, TypeScript, JavaScript, C, SQL
Frameworks: FastAPI, Node.js, React, Next.js, PyTorch, NumPy, pandas
Databases: PostgreSQL, Redis, DynamoDB
Cloud & DevOps: AWS (ECS, S3, CloudWatch), Terraform, Docker, GitHub Actions, Linux, OpenTelemetry, Prometheus, Grafana
Tools & Testing: Git, REST APIs, WebSockets, pytest, Vitest, Playwright